

# The Journey of Life

**II PUC · Ch 2 · Human Reproduction**

Starring Bindu 👧 & Prof. BAIO 🦉

A BAIO storybook by Narendra Sir · Biology Lecturer



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# Two Special Factories

Prof. BAIO opened a gentle science book. "In males, two **testes** sit in the **scrotum**, 2–2.5 °C cooler than the body. Inside are coiled **seminiferous tubules** that make sperms. **Leydig cells** make androgens, and **Sertoli cells** feed the developing sperms."

💡 REMEMBER

**Each testis has about 250 lobules with 1–3 seminiferous tubules each.**



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# The Sperm's Route

From the tubules, sperms travelled through the rete testis and vasa efferentia to the **epididymis**, then the vas deferens and the urethra. Glands — **seminal vesicles**, the **prostate** and bulbourethral glands — added fluid to make semen.

> 💡 **REMEMBER**
> **Male accessory glands secrete seminal plasma rich in fructose, calcium and enzymes.**



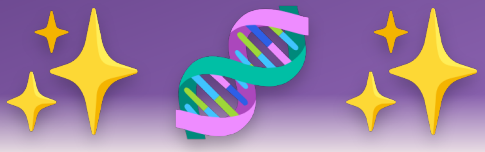


# The Female Palace

In females, two **ovaries** make ova and hormones. Each **fallopian tube** has a funnel-like infundibulum with finger-like fimbriae, a wider ampulla and a narrow isthmus. The uterus has three layers; its inner lining is the **endometrium**.

💡 **REMEMBER**

**Uterus wall: perimetrium (outer), myometrium (muscle), endometrium (glandular lining).**





# Making Gametes

**Spermatogenesis** starts at puberty: one spermatogonium gives four sperms. **Oogenesis** starts before birth: one oogonium gives one ovum and small polar bodies. The hormones **GnRH**, **LH** and **FSH** control it all.

💡 REMEMBER

**LH acts on Leydig cells; FSH acts on Sertoli cells.**






# The Monthly Calendar

The **menstrual cycle** lasts about 28 days. Menstruation is followed by the follicular phase. Around day 14, an **LH surge** causes **ovulation**. In the luteal phase, the **corpus luteum** makes progesterone to keep the endometrium ready.

💡 **REMEMBER**

**If there is no fertilisation, the corpus luteum degenerates and menstruation begins again.**



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# The Big Meeting

A sperm, with enzymes in its **acrosome**, reached the ovum at the **ampullary–isthmic junction**. Once one sperm touched the **zona pellucida**, the layer changed and blocked all the others. "The father's sperm, carrying X or Y, decides whether the baby is a girl or a boy," said Prof. BAIO.

💡 REMEMBER

**Fertilisation takes place at the ampullary–isthmic junction of the fallopian tube.**





# Finding a Home

The zygote divided — cleavage — into a **morula**, then a **blastocyst** with an outer **trophoblast** and an **inner cell mass**. The blastocyst settled into the endometrium — **implantation**. The **placenta** formed to feed the baby and make hormones like hCG and hPL.

💡 **REMEMBER**

**The inner cell mass forms the embryo; it contains stem cells.**



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# Hello, Baby!

After about nine months came birth — **parturition** — started by the foetal ejection reflex, which releases **oxytocin**. The mother's first milk, **colostrum**, carried IgA antibodies to protect the newborn. Bindu smiled: "What an amazing journey!"

💡 REMEMBER

**Relaxin from the ovary is secreted in the later phase of pregnancy.**



KEY TERMS

# Words you met in this story

Tap a word to jump to its page. Can you explain each one?

testes p.1 · scrotum p.1 · seminiferous tubules p.1 · Leydig cells p.1 · Sertoli cells p.1 · epididymis p.2 · seminal vesicles p.2 · prostate p.2 · ovaries p.3 · fallopian tube p.3 · endometrium p.3 · Spermatogenesis p.4 · Oogenesis p.4 · GnRH p.4 · LH p.4 · FSH p.4 · menstrual cycle p.5 · LH surge p.5 · ovulation p.5 · corpus luteum p.5 · acrosome p.6 · ampullary–isthmic junction p.6 · zona pellucida p.6 · morula p.7 · blastocyst p.7 · trophoblast p.7 · inner cell mass p.7 · implantation p.7 · placenta p.7 · parturition p.8 · oxytocin p.8 · colostrum p.8



QUICK QUIZ

# Do you remember?

**1. Androgens are secreted by:**

- Sertoli cells
- Leydig cells
- Spermatogonia
- Epididymis

**2. Fertilisation takes place in the:**

- Uterus
- Ampullary–isthmic junction
- Cervix
- Ovary

**3. Ovulation is triggered by a surge of:**

- FSH
- LH
- Progesterone
- Oxytocin

**4. The blastocyst implants into the:**

- Myometrium
- Endometrium
- Perimetrium
- Ovary

**5. Colostrum is rich in:**

- IgA
- IgE
- Insulin
- Oxytocin

Answers: 1-Leydig cells · 2-Ampullary–isthmic junction · 3-LH · 4-Endometrium · 5-IgA